CSC 163: Computing, Ethics, and Society
ABET SO4 · Module A2 (4b)

Copyright vs Patent Exercises Solution

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Exercise: Which right applies?

Name the main intellectual property right for each item: copyright, patent, trade secret, trademark, or none.

Item	Right
Source code of a mobile app	
A new video compression method described in claims of an issued USPTO patent	
Unreleased recommendation algorithm the company keeps confidential	
The app's name and logo	
Original icons and artwork in the interface	
The bare idea of 'a to-do list app'	

Exercise Solution: Which right applies?

Copyright: expression. Patent: claimed inventions. Trade secret: confidential know-how. Trademark: brand identifiers. Ideas alone: none.

Item	Right
Source code of a mobile app	**Copyright**
A new video compression method described in claims of an issued USPTO patent	**Patent**
Unreleased recommendation algorithm the company keeps confidential	**Trade secret**
The app's name and logo	**Trademark**
Original icons and artwork in the interface	**Copyright**
The bare idea of 'a to-do list app'	**None (an idea)**

Exercise: True or false

Write T or F. If false, correct it with a few words.

Statement	T / F	Correction (if false)
Independent creation is a defense to copyright infringement		
Independent creation is a defense to patent infringement		
Copyright protects an algorithm		
A patent lasts up to 20 years from the filing date		
Fair use is decided by a fixed formula		
To infringe a patent claim, a product must have every element of the claim (literally or by equivalent)		

Exercise Solution: True or false

1 point for T/F, 1 point for the correction.

Statement	T / F	Correction (if false)
Independent creation is a defense to copyright infringement	T	—
Independent creation is a defense to patent infringement	F	Patent infringement needs no copying; the product only has to meet the claim
Copyright protects an algorithm	F	It protects expression only; an algorithm may be patentable if eligible
A patent lasts up to 20 years from the filing date	T	—
Fair use is decided by a fixed formula	F	Courts weigh four factors case by case
To infringe a patent claim, a product must have every element of the claim (literally or by equivalent)	T	—

Exercise: Filtration (Altai)

In the abstraction-filtration-comparison test, unprotectable material is filtered out before comparison. For each element, decide whether it is filtered out or may be protected, and say why.

Element of the program	Filtered or protected?	Reason
The purpose: manage a store's inventory		
A routine written the only practical way to run on the hardware		
Menu layout required by the operating system's standard		
Code copied from a public-domain library		
Original, creative organization of modules with many alternatives		
Original artwork and icons		

Exercise Solution: Filtration (Altai)

Filtered out: ideas, efficiency (merger), external factors, public domain. Protected expression is what remains.

Element of the program	Filtered or protected?	Reason
The purpose: manage a store's inventory	**Filtered**	**An idea or function**
A routine written the only practical way to run on the hardware	**Filtered**	**Efficiency (merger)**
Menu layout required by the operating system's standard	**Filtered**	**External factors**
Code copied from a public-domain library	**Filtered**	**Public domain**
Original, creative organization of modules with many alternatives	**May be protected**	**Original expression**
Original artwork and icons	**May be protected**	**Original expression**

Exercise: Patent claim chart

A patent claim requires: (1) receiving a user's location; (2) retrieving nearby offers; (3) ranking offers by predicted interest; (4) displaying the top offers. Check each product against the claim. Does it literally infringe?

Product	**1**	**2**	**3**	**4**	**Literal infringement?**
X: does all four steps	Yes	Yes	Yes	Yes	
Y: lists offers alphabetically, no ranking	Yes	Yes	No	Yes	
Z: all four steps, built independently, team never saw the patent	Yes	Yes	Yes	Yes	

Exercise Solution: Patent claim chart

Literal infringement needs **all four elements**. Product Y lacks ranking, so there is no literal infringement (the doctrine of equivalents is a separate question). Product Z was built independently, but copying is not needed for patent infringement.

Product	1	2	3	4	Literal infringement?
X: does all four steps	Yes	Yes	Yes	Yes	**Yes: every element present**
Y: lists offers alphabetically, no ranking	Yes	Yes	No	Yes	**No: element 3 missing**
Z: all four steps, built independently, team never saw the patent	Yes	Yes	Yes	Yes	**Yes: independent creation is no defense**

Exercise: Practice 4b question

A competitor's photo app has a user interface, code organization and a filter method very similar to yours. Has it violated your **copyright** and/or **patent**? State the criteria you would apply to each right.

Exercise Solution: Practice 4b question

- **Model answer.**
- Copyright (UI, code organization): I must show a valid copyright, **copying** (access plus similarity), and substantial similarity in **protected** expression after filtering ideas, efficiency, standards and public domain. Defenses: fair use, license, independent creation.
- Patent (filter method): the method is not covered by copyright. Only a valid patent can protect it. I would compare each claim element to the competitor's app; infringement needs **every element**, and independent creation is no defense. The defendant may attack validity (prior art, Alice eligibility, obviousness).
- **Scoring (4 points):** both rights named and separated (1); copyright criteria (1); patent criteria (1); applied to the scenario with evidence and a conclusion or next step such as license or design-around (1).